

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			beta-carotene, chlorophyll and betalain and each one has only one spot on the chromatogram		1		1		
	(b)			award (1) for each correct answer chlorophyll is not present in carrot, tomato or beetroot extracts ☒ beta-carotene is present in carrot extract but not present in tomato extract ☐ both beta-carotene and betalain are present in beetroot extract ☒ betalain is present in tomato extract but not present in carrot extract ☐ both carrot and beetroot extracts contain a pigment other than beta-carotene, chlorophyll and betalain ☐ max (1) if three boxes ticked 0 if four or more boxes ticked			2	2		2
	(c)			0.55 (2) if answer incorrect award (1) for $\frac{4.4}{8}$	1	1		2	2	2
	(d)			one pigment in beetroot extract is insoluble (in the solvent used)		1		1		1
				Question 3 total	1	3	2	6	2	5